Acids and Bases Investigation

Is it an acid or a base?

Write down some things you know about acids and bases.

_____
_____
_____
_____
_____

Chemicals which are used to test whether substances are acids or bases are called _____.

Universal indicator undergoes several colour changes over a wide range of pH's. It can be used to determine the pH of a solution reasonably accurately. Which colours of universal indicator show: acidic solutions?

basic solutions?

neutral solutions?

Aim: To test a range of household products to determine whether they are acids or bases.

Equipment: A range of household substances
Universal indicator and colour chart
A depression tile
Safety glasses

Procedure:

1. Place a small amount of the substance to be tested in a dimple on the tile.
2. Add 3 drops of universal indicator to the substance.
3. Record the colour of the indicator.
4. Use the colour chart to determine the pH of the substance.
5. Classify each substance as an acid, a base or neutral.

Results:

Substance Tested	Colour of Indicator	pH	Acid/Base/ Neutral

Questions:

1. Strong acids have a pH of between 1 and 3. Which of the acids you tested are strong and which are weak?

2. Strong bases have a pH of between 12 and 14. Which of the bases you tested are strong and which are weak?

3. Which products are neutral?

4. Sally was cleaning her oven and she got some oven cleaner on her arm. She noticed it started to burn and itch. On the oven cleaner bottle it said the cleaner was a base. Which household product do you think would help stop the burning and why?